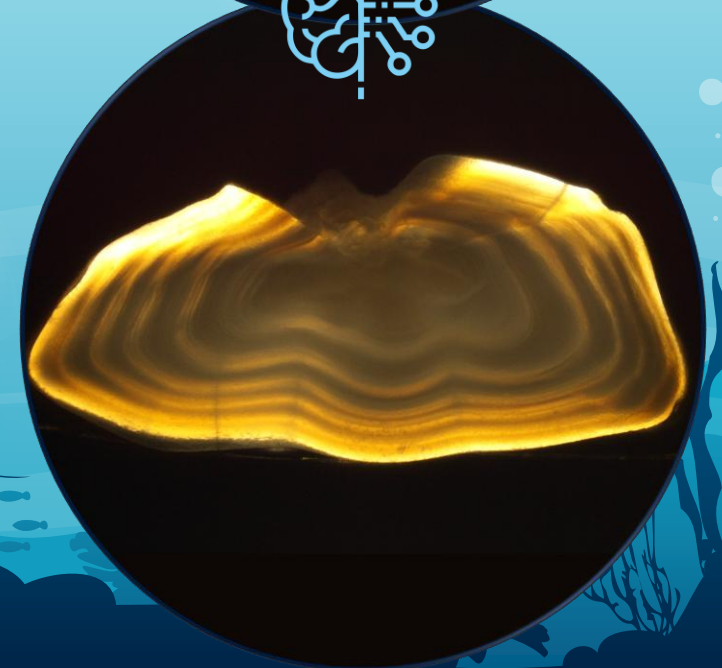
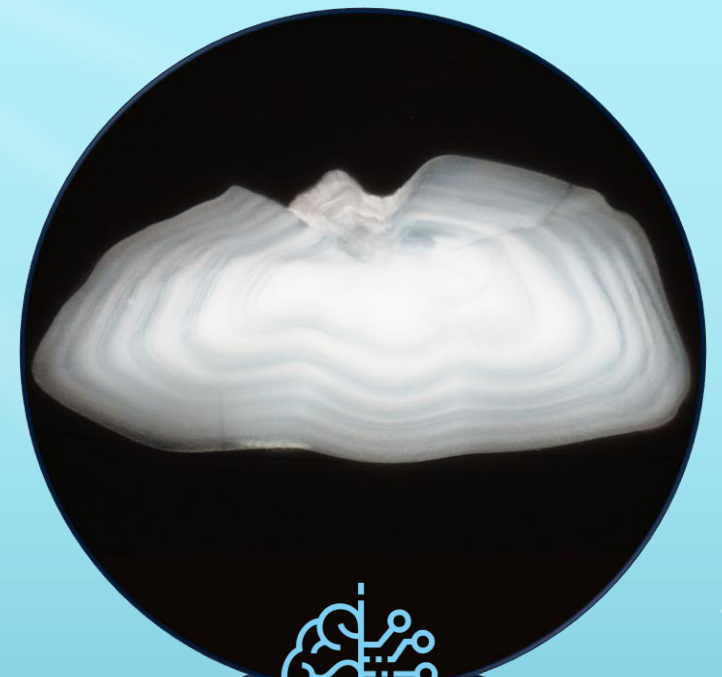


Automated fish aging beyond black boxes: perspectives from a pilot on the Northeast Atlantic saithe

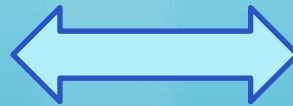
Côme Denechaud



Cayetano et al. 2024
Bojesen et al. 2024
Sigurðardóttir et al. 2023
Moen et al. 2023
Martinsen et al. 2022
Politikos et al. 2021
Ordoñez et al. 2020
Moen et al. 2018



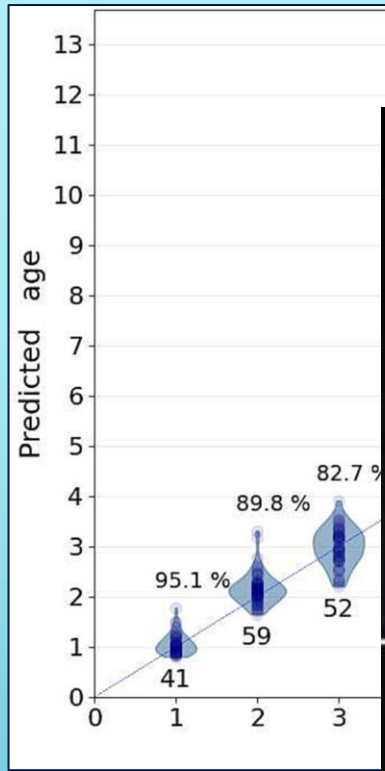
Explainable AI conundrum



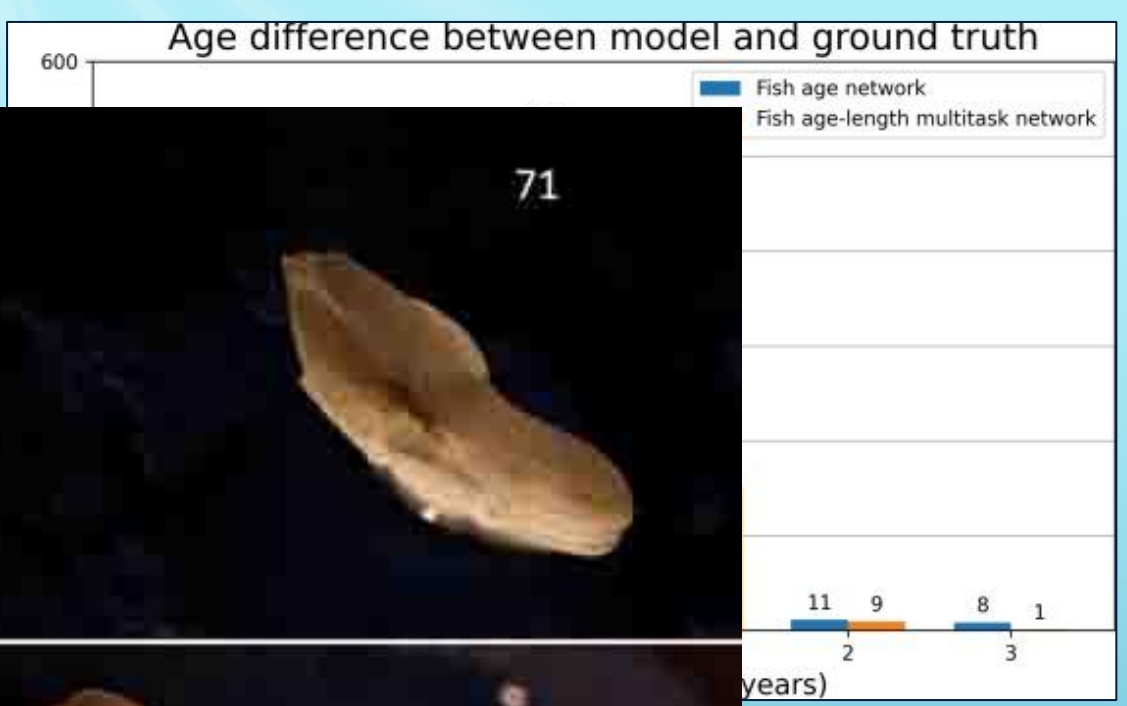
*Humans need a model of what the machine is doing,
the machine needs a model of what humans are doing*

Cayetano et al. 2024	Cod, Saithe, Haddock, Whiting	CNN (55-89) Mask-CNN (54-72) U-Net (54-72) ImgProc (26-54)
Bojesen et al. 2024	Cod	Autoregressive (69)
Sigurðardóttir et al. 2023	Plaice	CLIP (56) CNNs (45-54)
Moen et al. 2023	Cod	CNN ensembles (70-75)
Martinsen et al. 2022	Greenland halibut	CNN (24)
Politikos et al. 2021	Red mullet	CNN (64.4)
Ordoñez et al. 2020	Greenland halibut	CNN (25-30)
Moen et al. 2018	Greenland halibut	CNN (29)

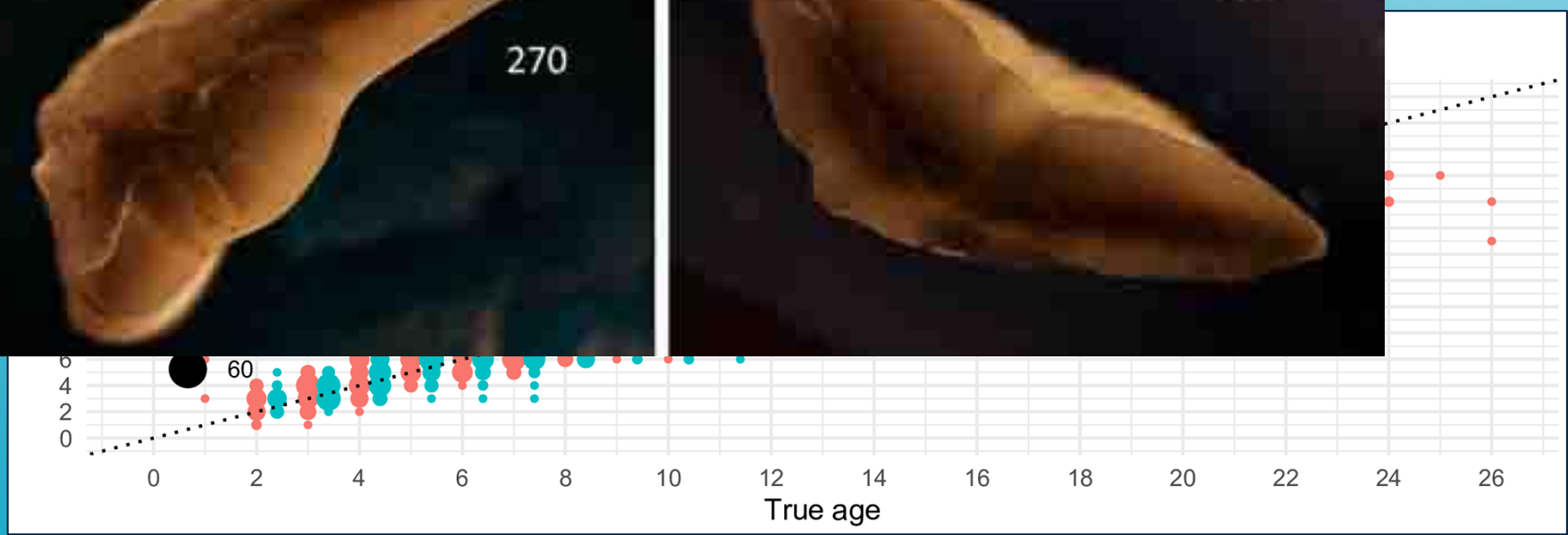




Moen et al. 2023



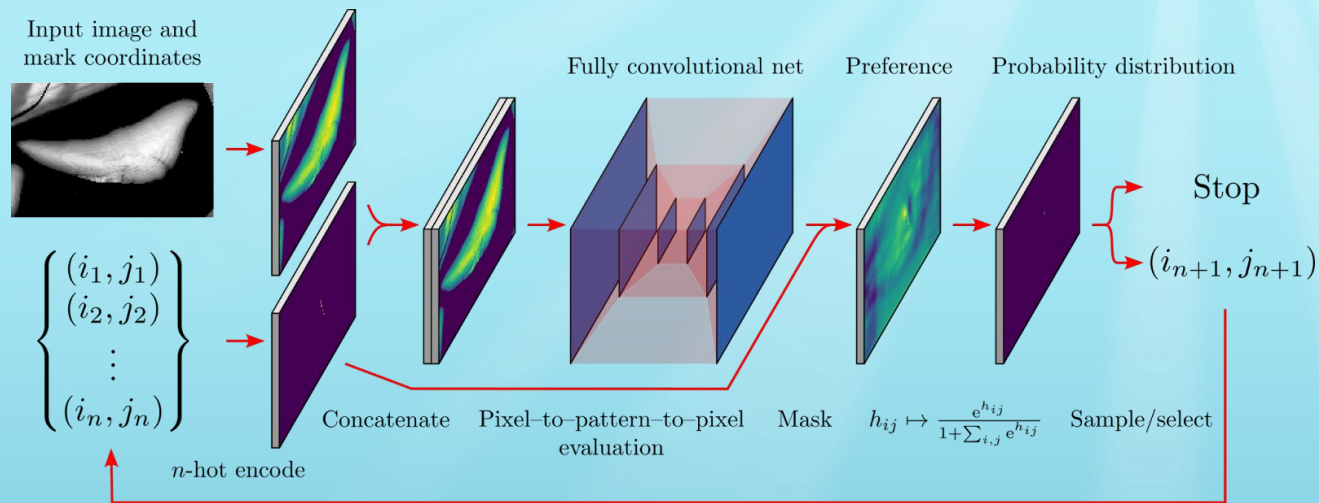
Politikos et al. 2021



Martinsen et al. 2022



Original concept



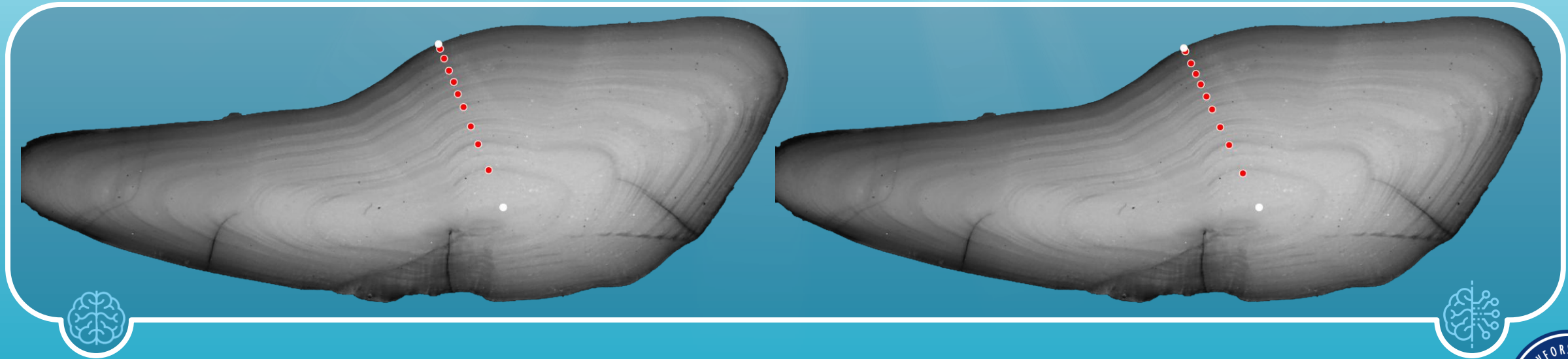
Annotating otoliths with a deep generative model

Troels Arnfred Bojesen¹, Côme Denechaud², Ketil Malde^{1,2*}

¹) Department of Informatics, University of Bergen, Norway

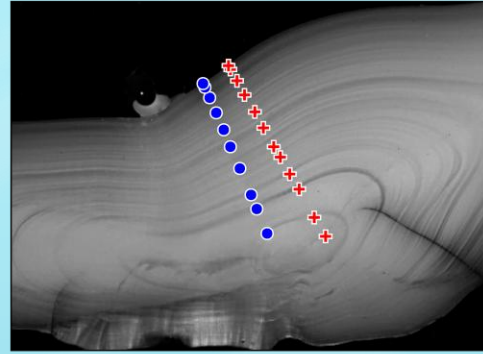
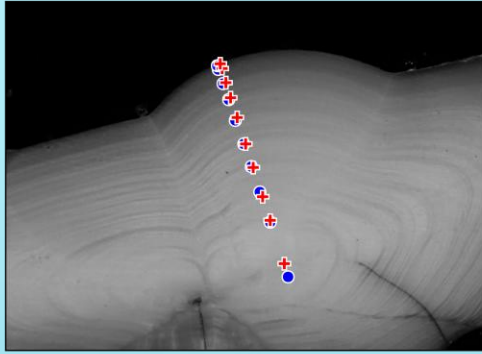
²) Institute of Marine Research, Bergen, Norway

*) Corresponding author: ketil.malde@imr.no

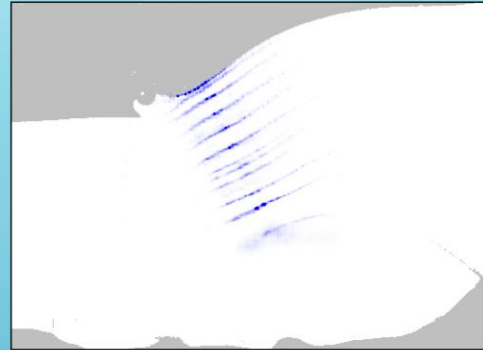
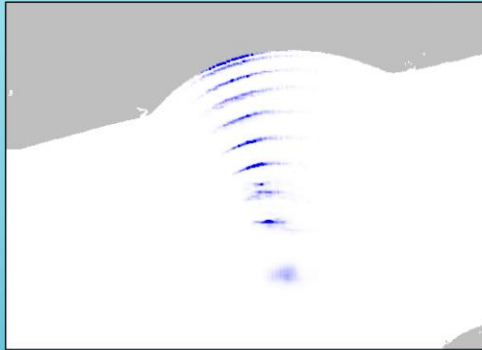


N-1938_45

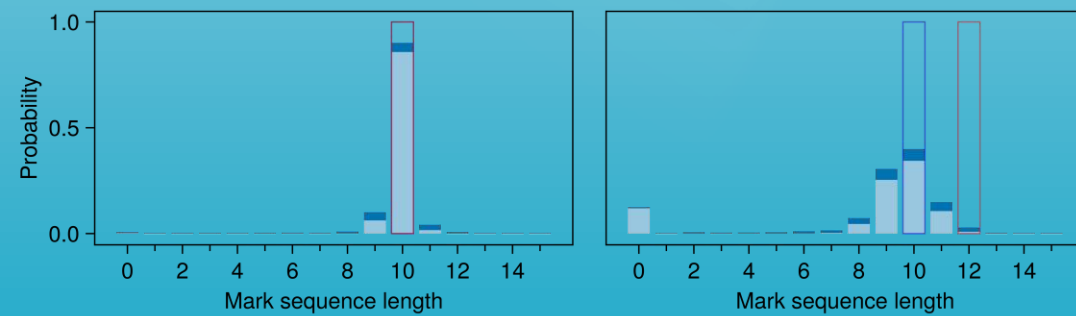
N-1948_05



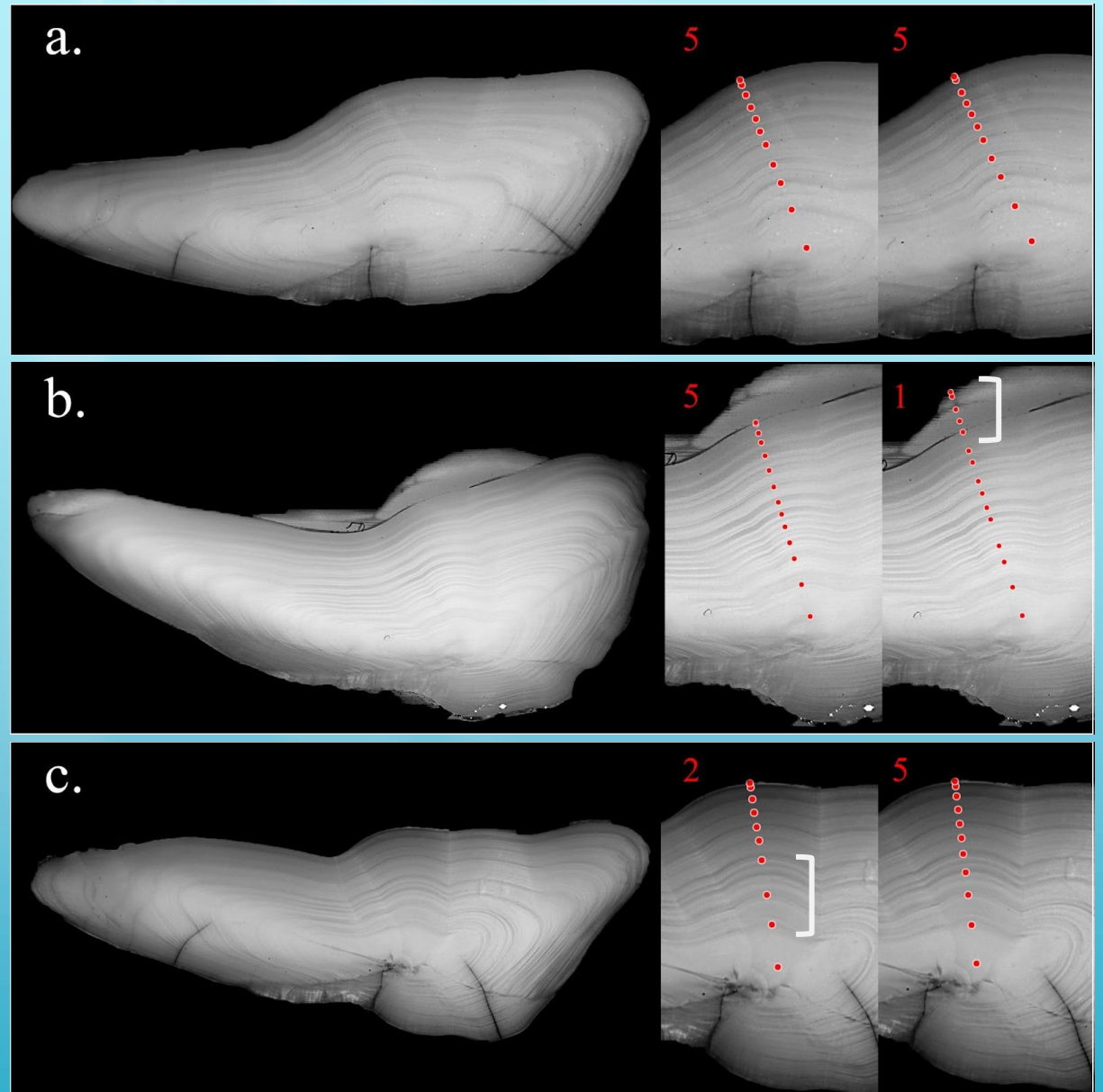
+ Human annotations • Greedy annotations



• Stochastic annotations



□ Human annotations □ Greedy annotations ■ Stochastic annotations



NEMO-framework



Biologiske data											
	Individ nr.	Vekt (kg)	Lengde (cm)	Kjønn	Stadium	Alder	Gytealder	Gytesoner	Lesbarhet	Otolitttype	Rand
1	1	0.925	49.500	1		4					
2	2	0.745	47.000	1		5					
3	3	0.725	45.500	2		4					
4	4	0.910	46.500	2		5					
5	5	0.325	40.500	1		4					
6	6	0.925	51.000	2		5					
7	7	0.565	42.000	2		4					
8	8	0.525	44.000	1		5					
9	9	0.460	43.000	1		5					
10	10	0.750	48.500	1		4					

	Alderslesningsnr.	Medium lest	Alder	Lesbarhet	Rand
1	1	2	5	1	4
2	2	1	5	1	4

Pros

- High output
- Quick turnover
- No lab cost

Cons

- Limited QA
- Unknown temporal stability
- «Wasted» potential

NEMO-framework

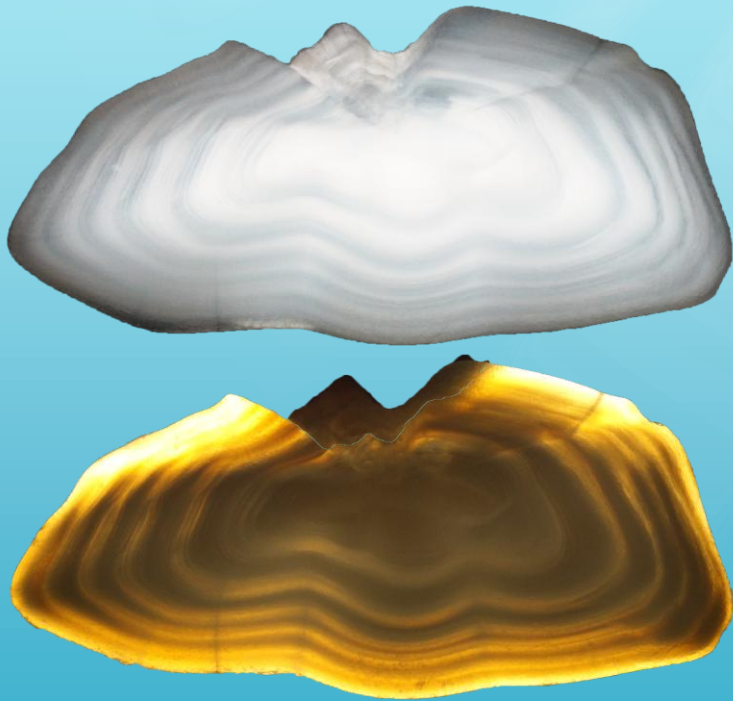
2 «entry» species: **saithe** (*Pollachius virens*) & capelin (*Mallotus villosus*)

Goal: ~ 5000 otoliths per year from 2024-2026

Sources: both survey and commercial sampling

Coverage: entire range from BS to NS

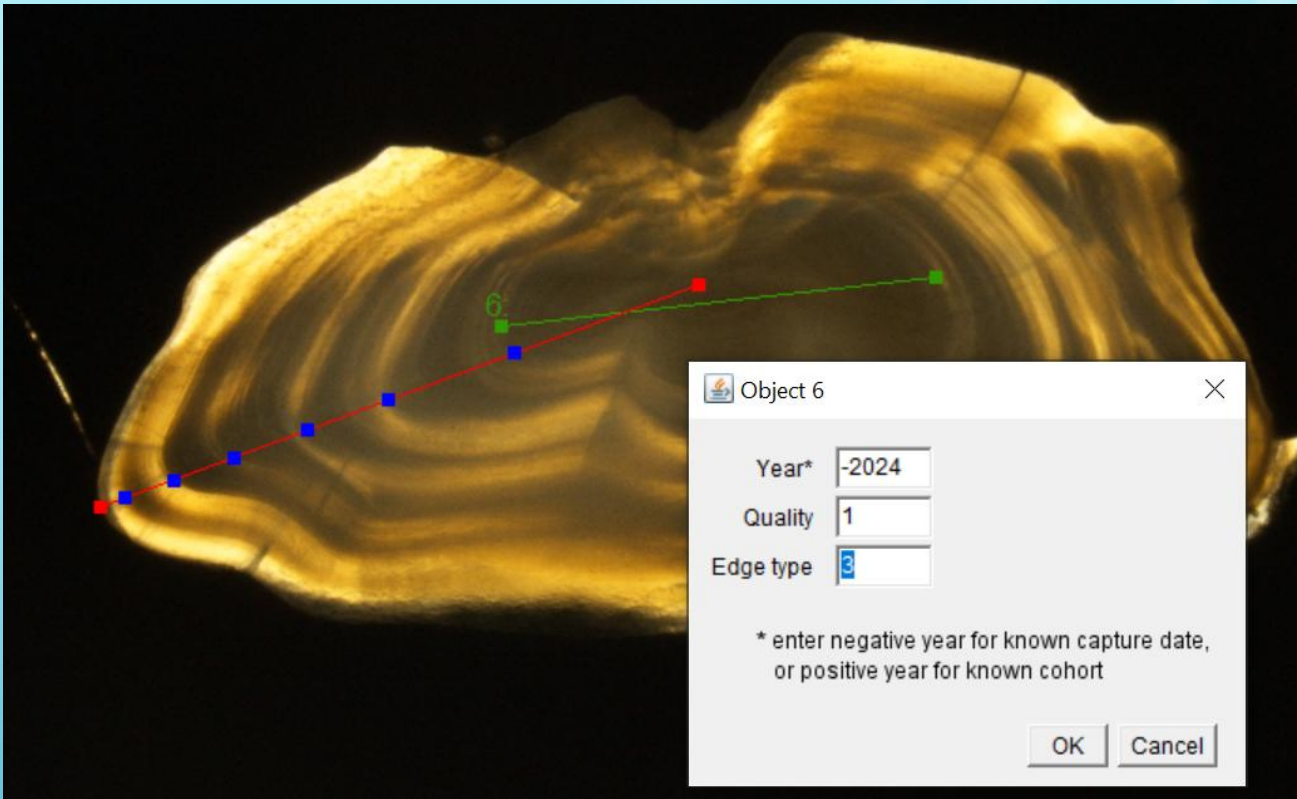
Process: embedded, thin-sectioned, imaged and annotated



- Standard otolith ; framing ; expert ; reading axis
- Non-destructive annotation in OA plugin



NEMO-framework



Sample	Quality	Age	Edge	Cohort	Core	I	Year	Increment	X	Y
Overlys_1	1	5	4	2019	1645.2	core			1230	1113
Overlys_1	1	5	4	2019	1645.2	1	2020	514.1	1113	1263
Overlys_1	1	5	4	2019	1645.2	2	2021	320.1	1041	1356
Overlys_1	1	5	4	2019	1645.2	3	2022	273.1	978	1435
Overlys_1	1	5	4	2019	1645.2	4	2023	114.1	953	1468
Overlys_1	1	5	4	2019	1645.2	5	2024	139.7	921	1509
Overlys_1	1	5	4	2019	1645.2	edge		52.1	909	1524

Otolitt_annotering.ibtsq1_janicke.oj

Images Objects Columns Qualifiers

Object.J Results Show in Explorer

☒ Show Object Layer ☐ Virtual Flag

Linked Images	Objects	Stack size	px/unit
2024_60024_01.tif	1 (1 - 1)	1	-
2024_60024_02.tif	0	1	-
2024_60024_03.tif	0	1	-
2024_60024_04.tif	0	1	-
2024_60024_05.tif	0	1	-
2024_60024_06.tif	0	1	-
2024_60024_07.tif	0	1	-
cal 1.7x Nikon SMZ25 SHR...	0	1	0.3600

Link... Link all Unlink

Scale Linked Images

Current Image:
cal 1,7x Nikon SMZ25 SHR plan Apo 1x WD60.tif

New Scale 0.360 px/unit

Unit um

☒ Propagate to all linked images

Change Scale Cancel



Development

Collection + training ML

Ages trad. + image

Assessment as usual

2024-2025

Pilot

Desired ML «behavior»

Ages trad. + image

Test assessment with each

2026

Routine

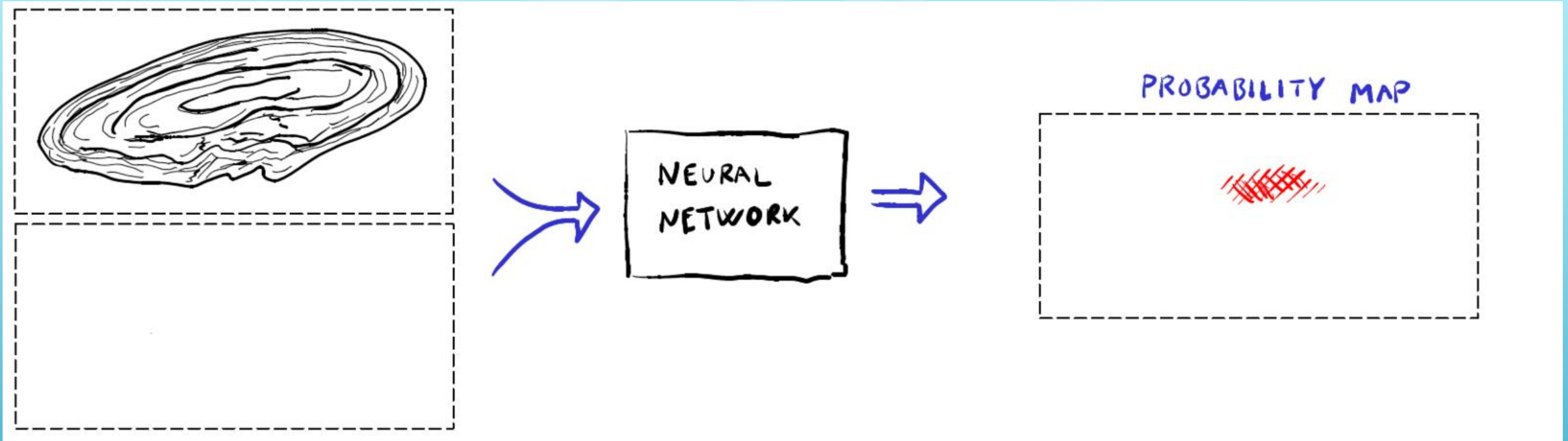
% read only by ML

Continuous QA + ML dev.

Assessment uses both

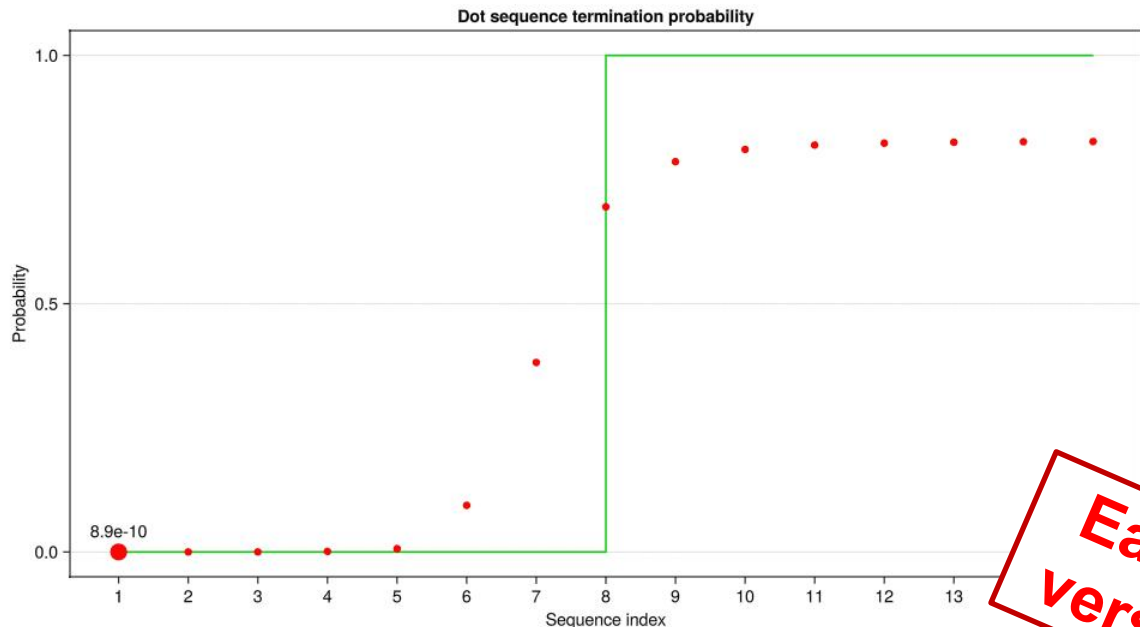
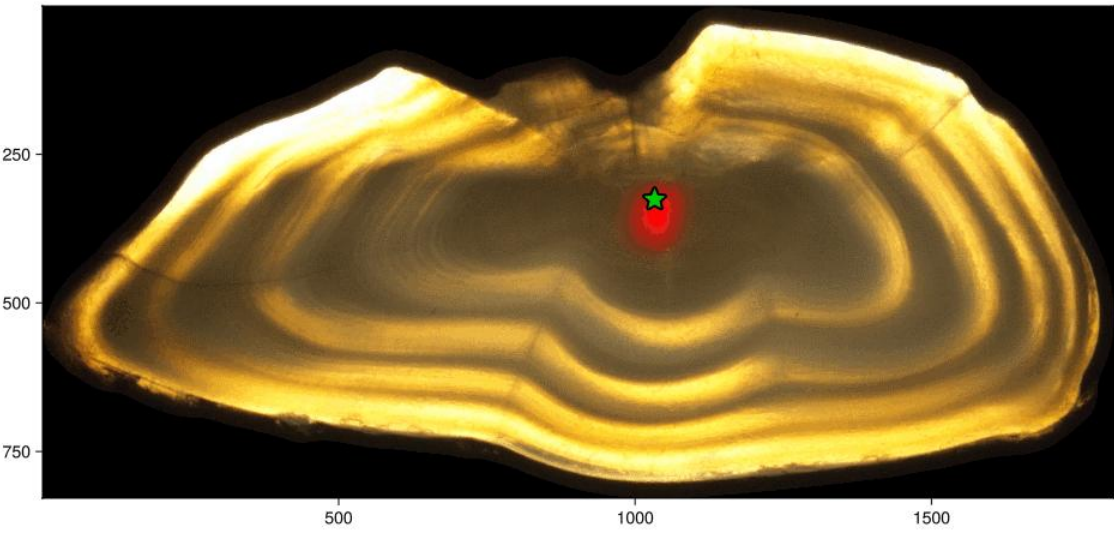
2027-...

Pilot project

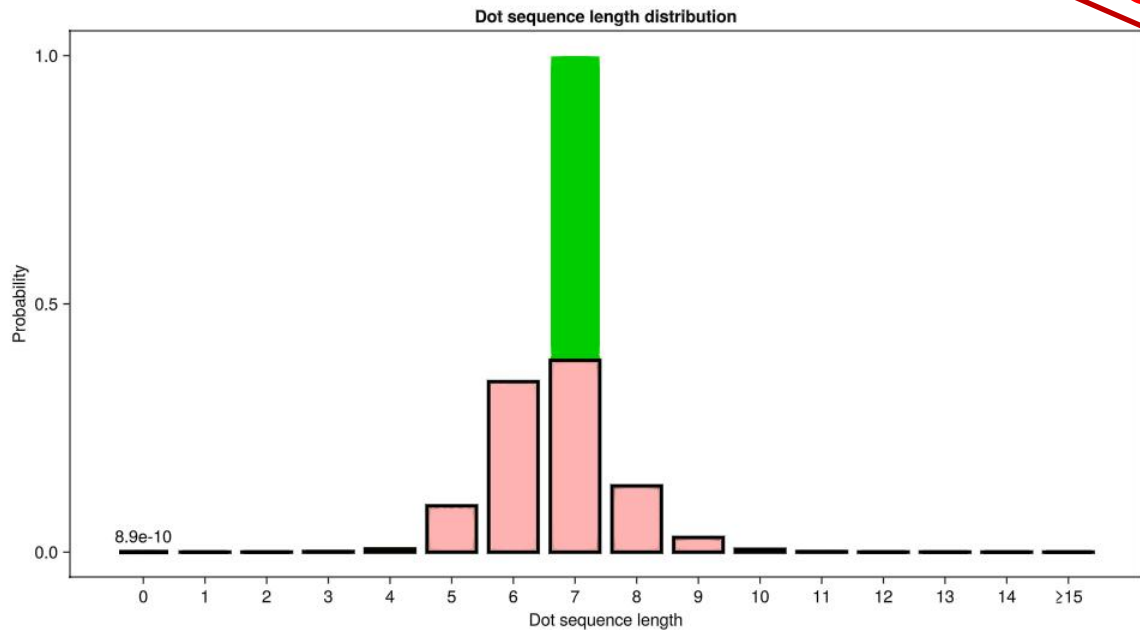
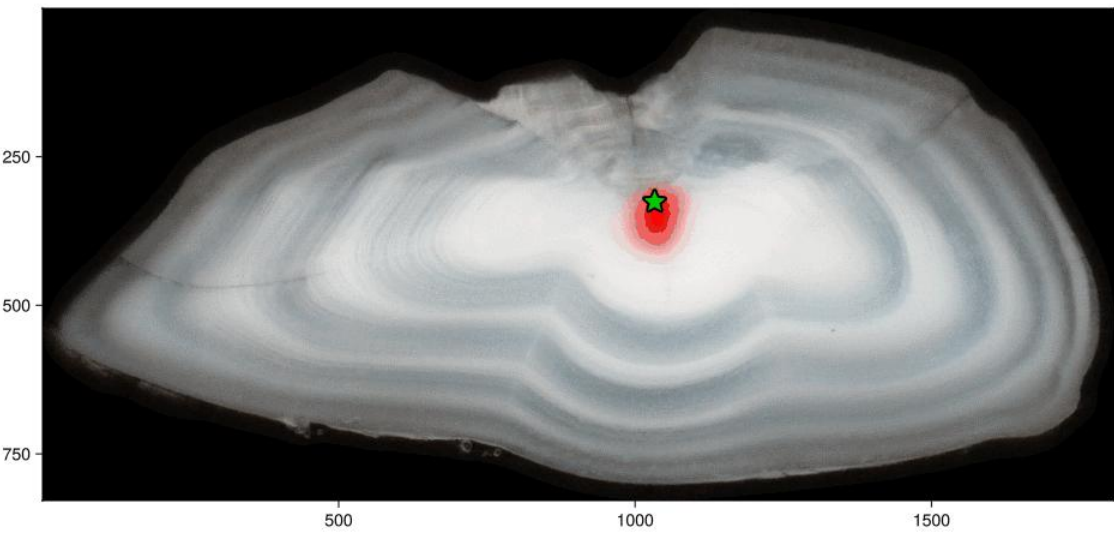


Pilot project

Sample 2024_60251_07



Earlier version



Pilot project

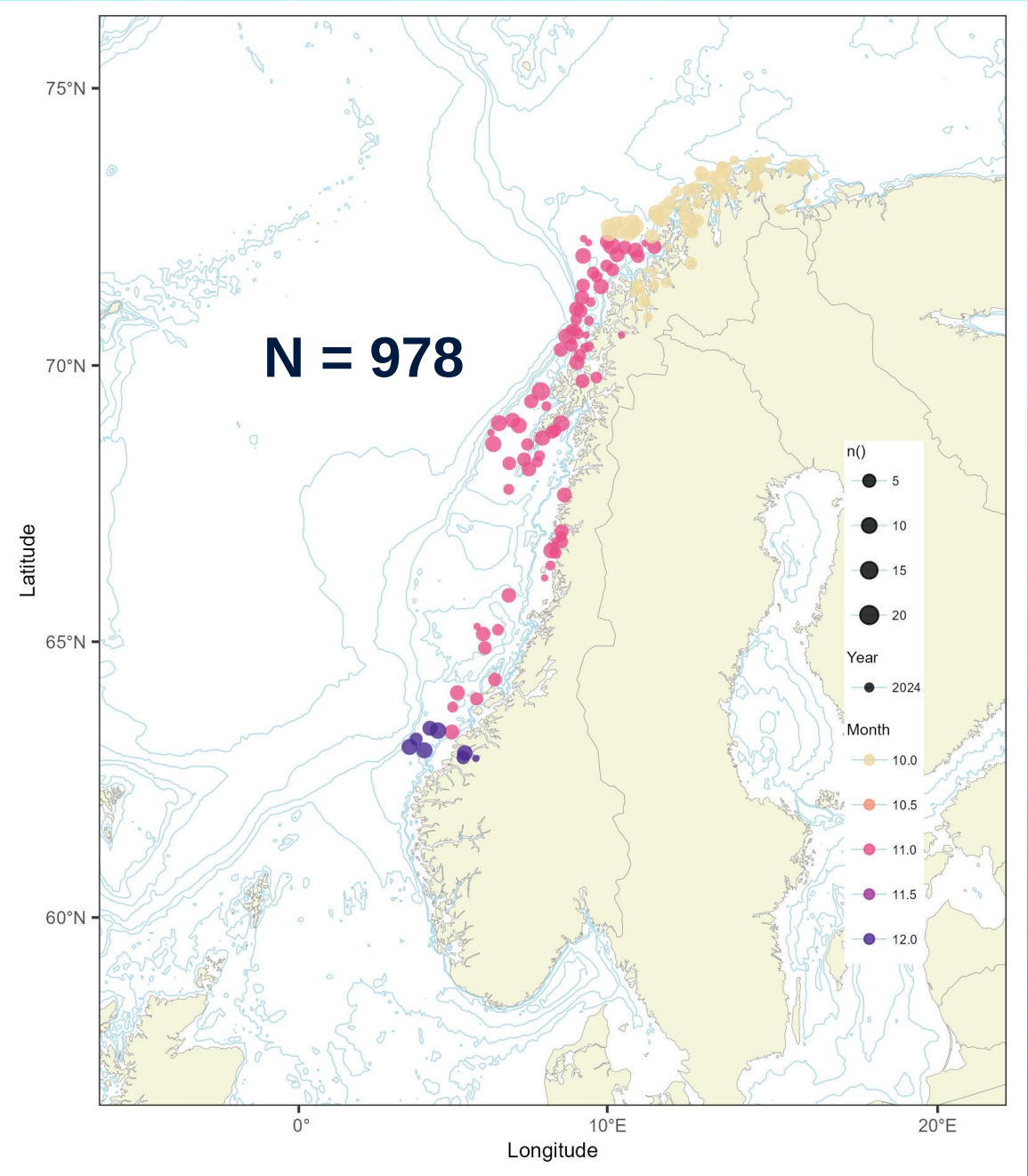
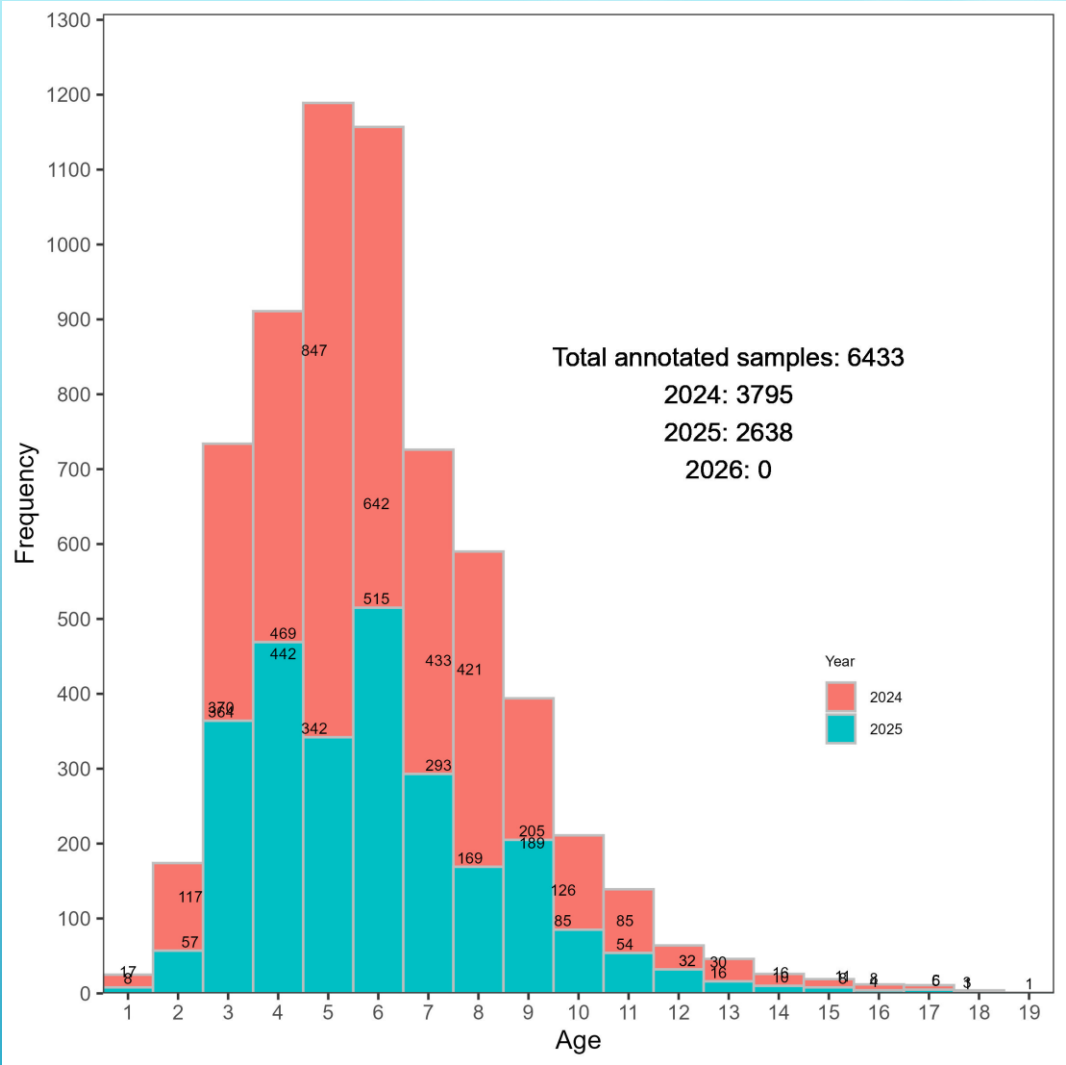
Category 1 stock

- Age-based
- **Annual survey index**
- Commercial catches & samples



Autumn coastal survey
Acoustic + catch tuning
Delta-GAM index

Pilot project



Pilot project

Category 1 stock

- Age-based
- ~~Annual survey index~~
- Commercial catches & samples

Autumn coastal survey
Acoustic + catch tuning
Delta-GAM index

↓ ML

Performance

- Test effect on index
- Test AEM

Validity

- Compare readings
- Measure performance

Fish	1	2	3	...	8	9	10	Exp.
Indiv. 1	0	0	0	...	0.84	0.14	0.03	8
Indiv. 2	0	0	0	...	0.56	0.44	0	9
Indiv. 3	0.20	0.65	0.15	...	0	0	0	3

Pilot project

Category 1 stock

- Age-based
- **Annual survey index**
- Commercial catches & samples

Performance

- Test different proportions
- Test AEM

Validity

- Compare readings
- Measure performance

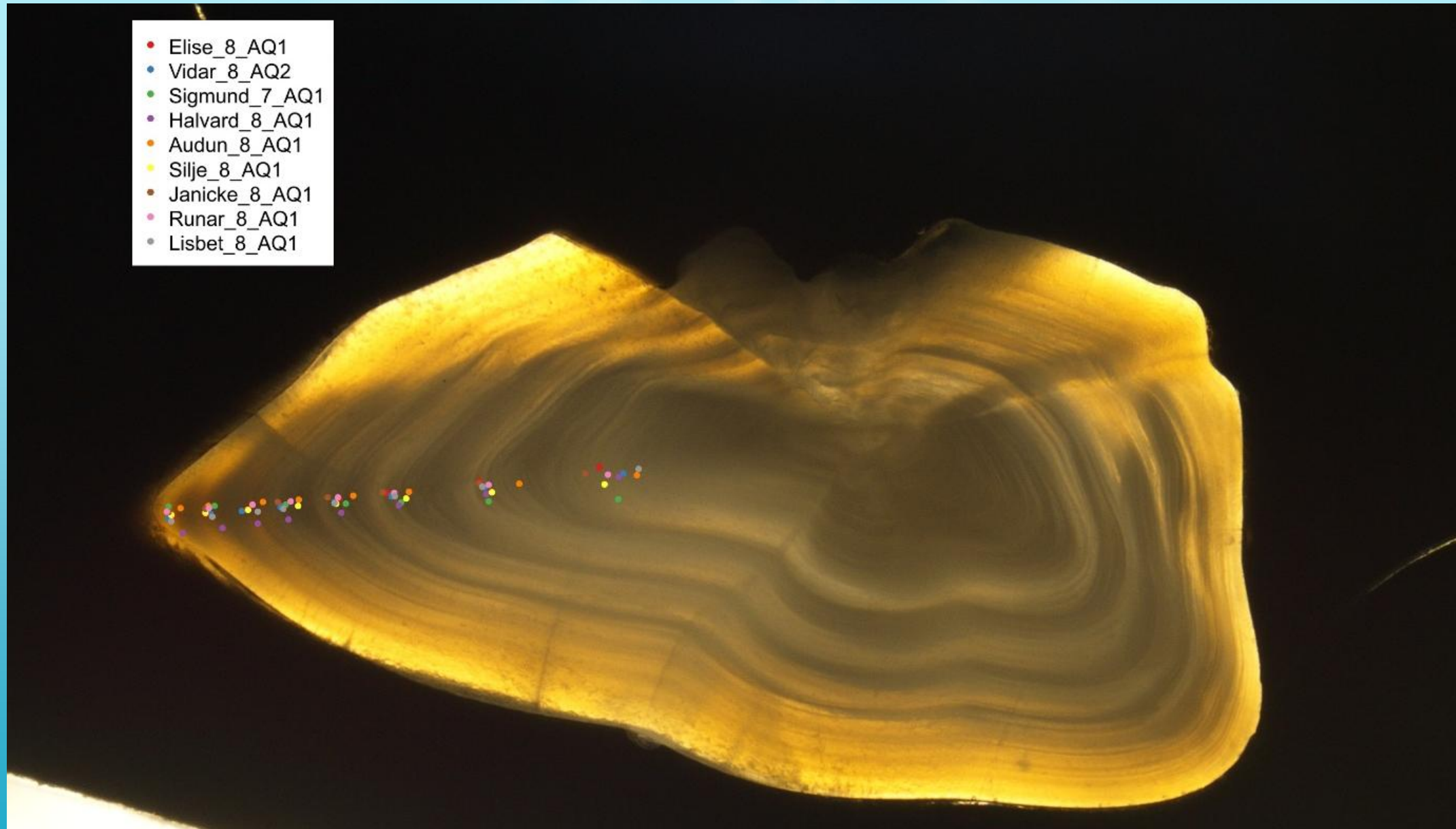
Cross-reading agreement

- ➔ 3 estimates per sample
- ➔ Trad vs annotated vs ML

Multi-reader performance

- ➔ 100 samples, 9 readers
- ➔ The 10th Dentist?

Pilot project



Pilot project

Category 1 stock

- Age-based
- **Annual survey index**
- Commercial catches & samples

Performance

- Test different proportions
- Test AEM

Validity

- Compare readings
- Measure performance

Cross-reading agreement

- ➔ 3 estimates per sample
- ➔ Trad vs annotated vs ML

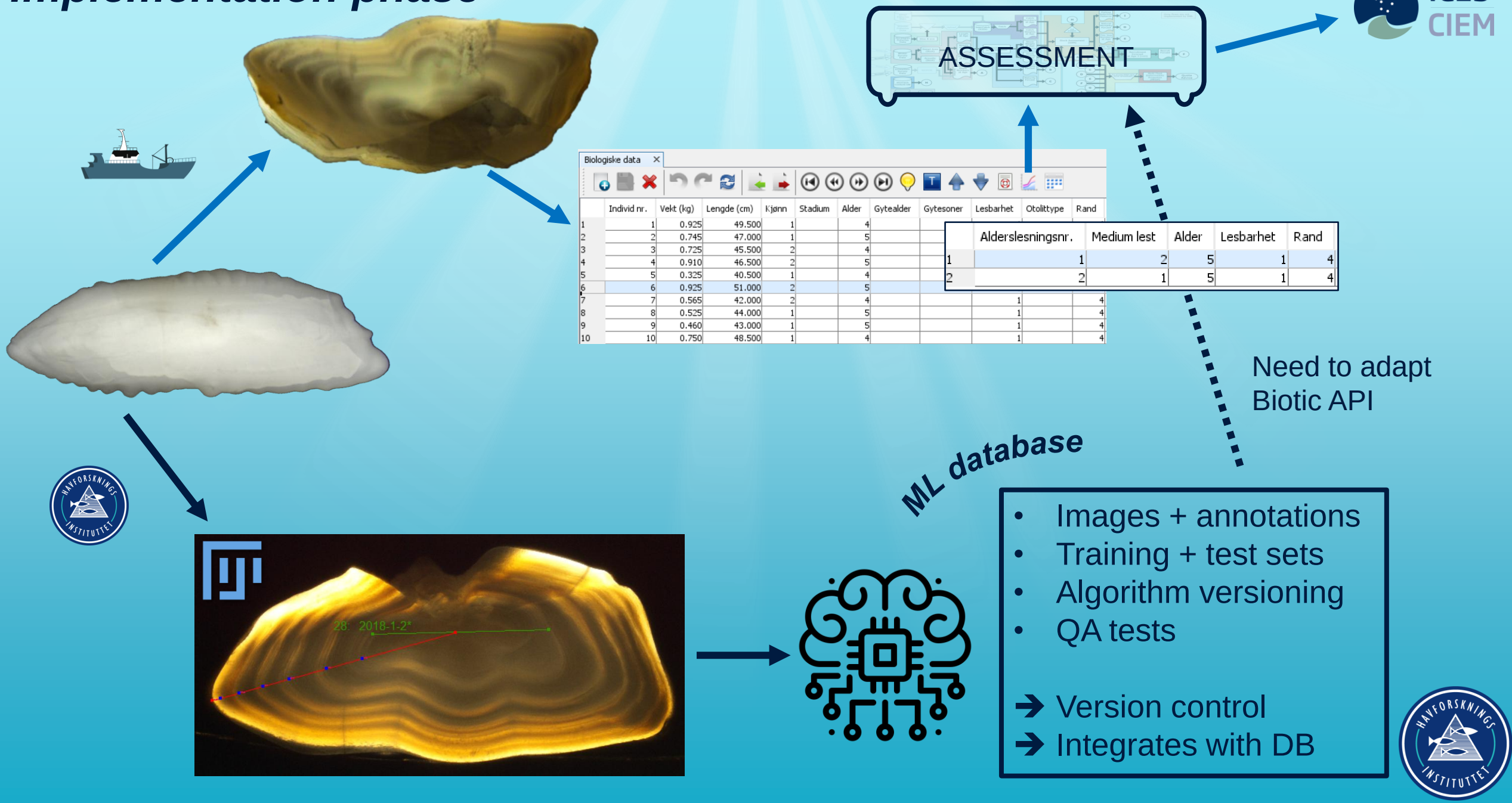
Multi-reader performance

- ➔ 100 samples, 9 readers
- ➔ The 10th Dentist?

Lab. comparison

- ➔ Same output, different setup
- ➔ Over 800 unknown images

Implementation phase



Implementation phase

Method	Preparation	Imaging	Aging time	Weekly output
Traditional				
Machine				



Implementation phase

Method	Preparation	Imaging	Aging time	Weekly output
Traditional	Negligible			
Machine	1000-1250 / week			



Implementation phase

Method	Preparation	Imaging	Aging time	Weekly output
Traditional	Negligible	None		
Machine	1000-1250 / week	~700 / week ↓ ~2000 week		



Implementation phase

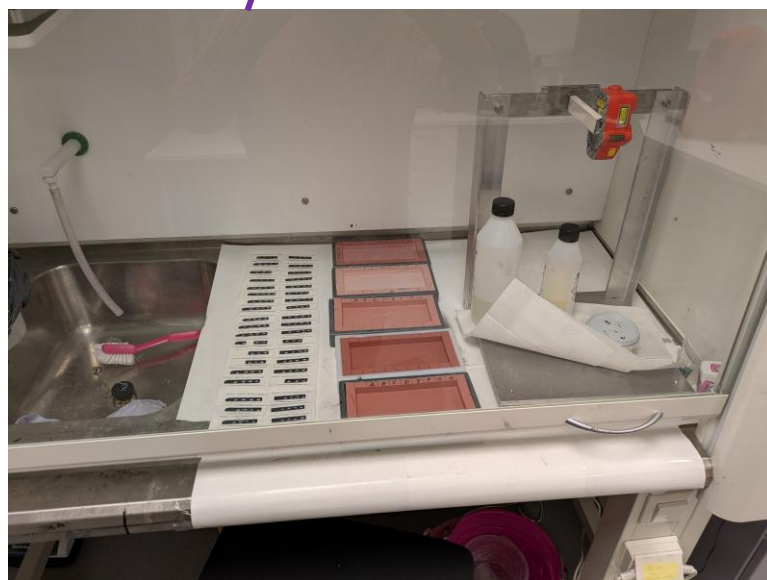
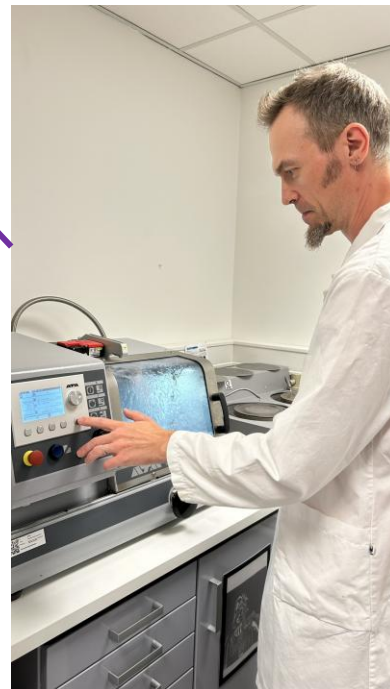
Method	Preparation	Imaging	Aging time	Weekly output
Traditional	Negligible	None	30sec – 3min	
Machine	1000-1250 / week	~700 / week ↓ ~2000 week	< 5 sec	

Implementation phase

Method	Preparation	Imaging	Aging time	Weekly output
Traditional	Negligible	None	30sec – 3min	>2000 (opt) 500-800 (real)
Machine	1000-1250 / week	~700 / week ↓ ~2000 week	< 5 sec	~700 (now) > ...?

Perspectives

- Likely cost efficient
- Concrete output for validation and uncertainty
- Higher scientific output (growth measurements, etc)
- High transfer learning potential



Thanks!

